

AI-POWERED PHISHING & BEC INVESTIGATION PLATFORM

Project Report

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GitHub: github.com/PratikGulig/AI-Powered-Phishing-BEC-Investigation-Platform

Technology: Python | Streamlit | Claude AI | VirusTotal API | MITRE ATT&CK

Date: June 2026

1. Executive Summary

The AI-Powered Phishing & BEC Investigation Platform is a comprehensive cybersecurity tool designed to detect, analyze, and report on phishing emails and Business Email Compromise (BEC) attacks. Built using Python and Streamlit, the platform integrates multiple threat intelligence sources and AI-driven analysis to provide security teams with actionable intelligence in real time.

The tool accepts a raw email body as input and performs a multi-layered analysis including AI-based risk scoring, BEC pattern detection, domain age verification, Virus Total URL scanning, and MITRE ATT&CK technique mapping. Results are presented in a structured dashboard and can be exported in multiple formats including PDF, JSON, CSV, and Wazuh-compatible alert formats.

2. Project Overview

Project Name	AI-Powered Phishing & BEC Investigation Platform
Developer	Pratik Gulig
GitHub Repository	github.com/PratikGulig/AI-Powered-Phishing-BEC-Investigation-Platform
Frontend Framework	Streamlit (Python)
AI Model	Claude AI (Anthropic)
Port / URL	localhost:8501
Date	June 2026
Category	Cybersecurity / Threat Intelligence / Blue Team

3. Problem Statement

Phishing and Business Email Compromise (BEC) attacks remain among the most financially devastating cyber threats globally. According to the FBI IC3 Report, BEC attacks have caused over \$12 billion in reported losses. Traditional email security tools often fail to provide:

- Deep contextual AI analysis of email intent and language patterns
- Real-time URL reputation checking against threat intelligence databases
- Automated mapping of attack techniques to the MITRE ATT&CK framework
- Exportable forensic artifacts for SIEM and incident response integration

This project addresses these gaps by building an all-in-one email investigation platform that combines AI reasoning with structured threat intelligence.

4. Project Objectives

- Analyze suspicious emails for phishing indicators using rule-based and AI-driven approaches
- Detect Business Email Compromise (BEC) patterns including urgency cues and financial keywords
- Perform automated domain age and WHOIS lookups to identify newly registered malicious domains
- Integrate with VirusTotal API for URL reputation and malware detection
- Map detected attack patterns to MITRE ATT&CK techniques automatically
- Generate a human-readable AI explanation of why an email is suspicious
- Provide downloadable forensic reports in multiple formats (PDF, JSON, CSV, Wazuh Alert)

5. Technology Stack

Component	Technology / Tool
Programming Language	Python 3.x
Web Framework	Streamlit
AI Engine	Claude AI (Anthropic API)
URL Threat Intelligence	VirusTotal API v3
Domain Intelligence	WHOIS Lookup (python-whois)
Attack Framework	MITRE ATT&CK Framework
Report Generation	FPDF2 (PDF), JSON, CSV
SIEM Integration	Wazuh Alert Export
Deployment	Local (localhost:8501)

6. Key Features

6.1 Risk Scoring Engine

The platform calculates a 0–100 phishing risk score based on a weighted combination of heuristic rules including suspicious domain patterns, urgency language, financial keywords, and URL characteristics. Scores are classified into severity tiers: Low, Medium, High, and Critical.

6.2 BEC (Business Email Compromise) Detection

The tool specifically scans for BEC indicators such as:

- Urgency-inducing keywords (e.g., 'immediately', 'verify now', 'urgent')
- Financial trigger words (e.g., 'funds', 'transfer', 'wallet', 'bank account')
- Impersonation patterns (executive or vendor impersonation)

When BEC indicators are found, a prominent alert is displayed to the analyst.

6.3 Domain Age & WHOIS Lookup

The platform extracts all URLs from the email body and performs WHOIS lookups to determine domain registration age. Newly registered domains (common in phishing campaigns) are flagged as high risk.

6.4 Virus Total URL Analysis

Each extracted URL is submitted to the virus Total API to retrieve community-sourced malware, phishing, and suspicious detection counts. Results are displayed with Malicious, Suspicious, and Harmless scores.

6.5 MITRE ATT&CK Mapping

The tool automatically maps detected behaviours to the MITRE ATT&CK framework, providing standardized technique identifiers such as:

- T1566.002 — Spear phishing Link
- T1056 — Input Capture
- T1656 — Impersonation
- T1566 — Phishing

6.6 AI-Powered Explanation (Claude AI)

Using the Anthropic Claude API, the platform generates a structured natural-language explanation of the analysis results, including why the email is suspicious, specific indicators observed, and recommended actions for the recipient.

6.7 Multi-Format Export

Analysts can download investigation results in four formats:

- PDF Report — formatted investigation summary
- IOC JSON — machine-readable indicators of compromise
- IOC CSV — spreadsheet-compatible IOC list
- Wazuh Alert — pre-formatted JSON alert for Wazuh SIEM ingestion

7. System Architecture

The platform follows a modular pipeline architecture:

INPUT LAYER

Raw Email Text (Paste via Streamlit UI)

ANALYSIS LAYER

1. Header & Sender Parsing
2. Heuristic Risk Scoring (0-100)
3. BEC Pattern Detection
4. URL Extraction & WHOIS Domain Age
5. VirusTotal API Scan
6. MITRE ATT&CK Mapping
7. Claude AI Explanation Generation

OUTPUT LAYER

Dashboard: Risk Score, Severity, BEC Alert, MITRE ATT&CK, AI Explanation
Downloads: PDF | IOC JSON | IOC CSV | Wazuh Alert

8. Results & Screenshots

The following screenshots demonstrate the platform in action, analyzing a real-world phishing email sample impersonating Binance.

8.1 Application Interface

The main interface provides a clean text input area where analysts can paste the full email body for investigation.

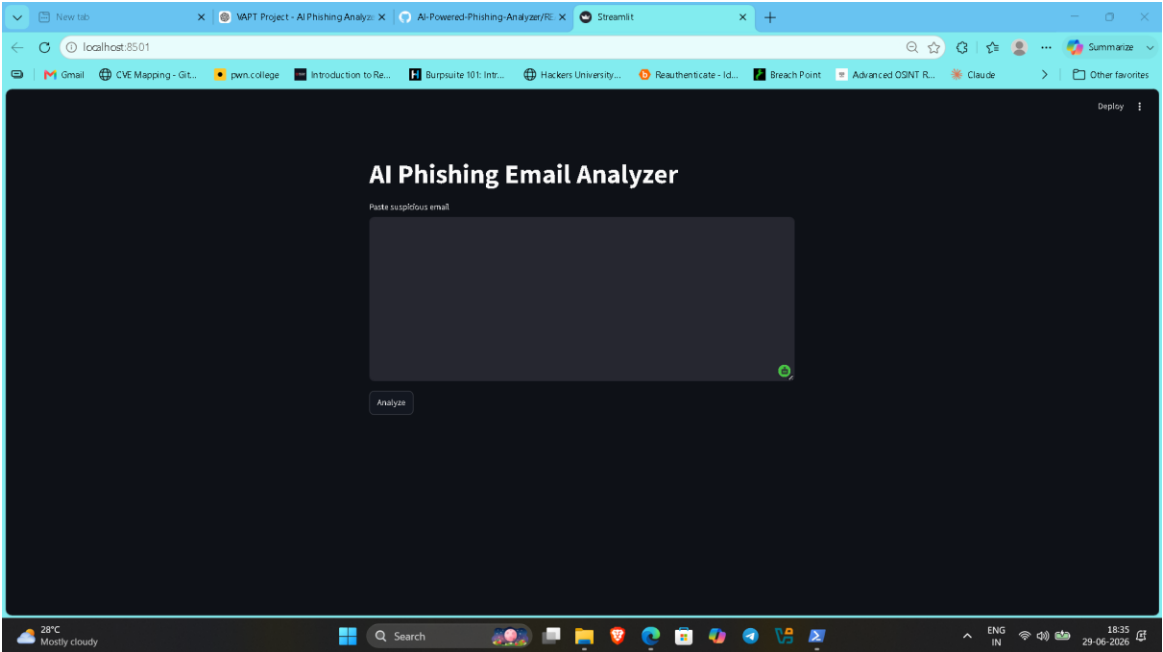


Figure 1: Main application interface — AI Phishing Email Analyzer on localhost:8501

8.2 Risk Score & Severity Results

After clicking Analyze, the platform displays the computed Risk Score and Severity classification. The sample phishing email (impersonating Binance with a wallet suspension lure) received a Risk Score of 85 with a HIGH severity rating. The identified sender email address (support@binance-security.net) is also displayed.

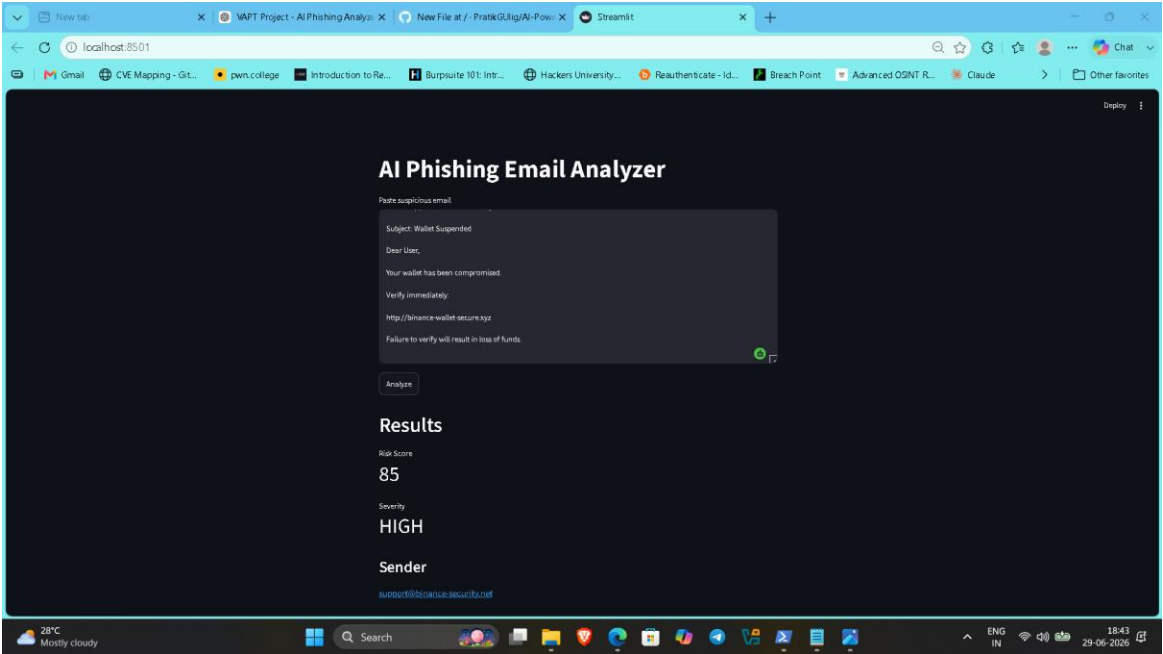


Figure 2: Analysis results showing Risk Score 85, Severity HIGH, and identified sender

8.3 BEC Analysis, Domain Age & VirusTotal Results

The platform detected Business Email Compromise patterns in the email, flagging urgency language ('immediately') and financial keywords ('funds'). The domain 'binance-wallet-secure.xyz' was identified as the malicious URL. WHOIS lookup was performed and VirusTotal scan results were retrieved for the extracted URL.

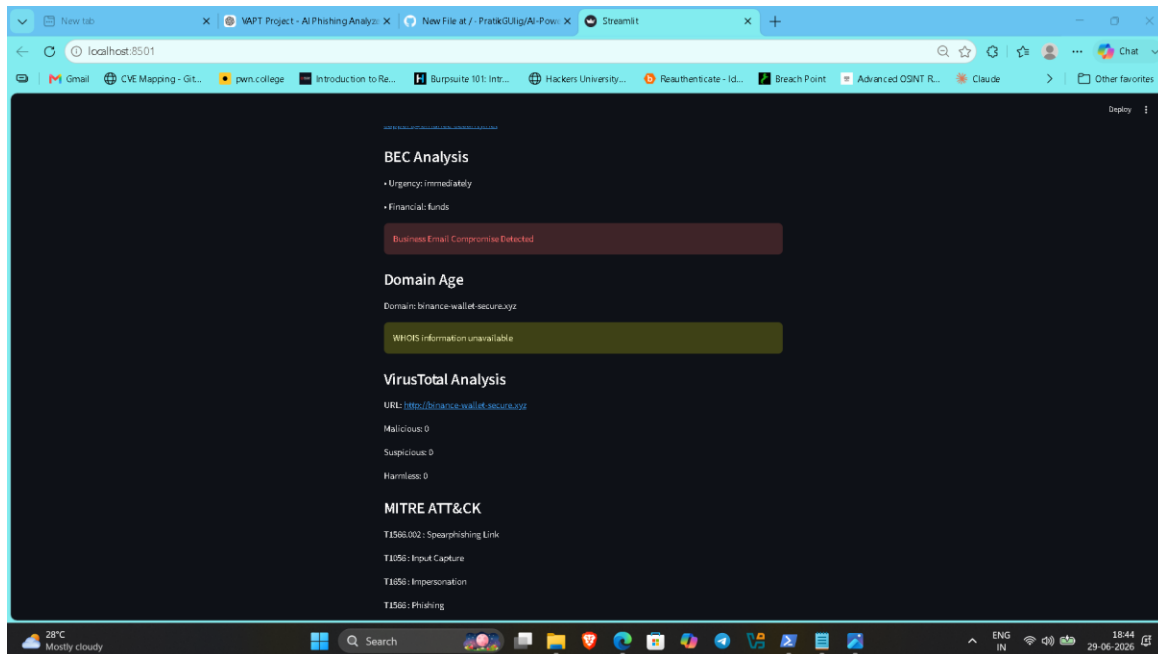


Figure 3: BEC Detection, Domain Age (WHOIS), VirusTotal Analysis, and MITRE ATT&CK mapping

8.4 AI Explanation & Export Options

Claude AI generated a detailed natural-language analysis explaining why the email is suspicious, what specific red flags were detected, and what actions the recipient should take. Export options are provided for PDF, IOC JSON, IOC CSV, and Wazuh Alert formats.

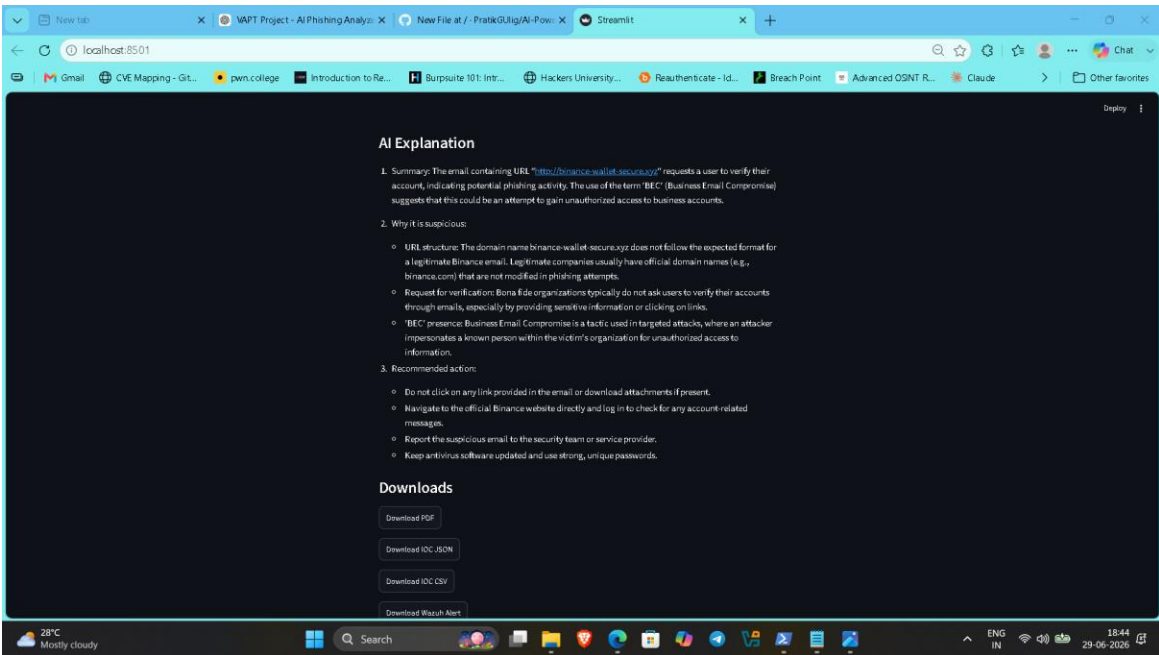


Figure 4: AI-generated explanation and download options for forensic report export

9. Sample Analysis Walkthrough

The following phishing email sample was used to validate the platform:

Subject: Wallet Suspended

Dear User,

Your wallet has been compromised.
Verify immediately:
<http://binance-wallet-secure.xyz>
Failure to verify will result in loss of funds.

Analysis Results Summary

Risk Score	85 / 100
Severity	HIGH
Sender	support@binance-security.net
Malicious URL	http://binance-wallet-secure.xyz
BEC Indicators	Urgency: 'immediately' Financial: 'funds'
BEC Alert	Business Email Compromise Detected

WHOIS Domain Age	WHOIS information unavailable (new domain)
VirusTotal	Malicious: 0 Suspicious: 0 Harmless: 0
MITRE ATT&CK	T1566.002, T1056, T1656, T1566
AI Verdict	High-confidence phishing / BEC attempt

10. Indicators of Compromise (IOC) Export

The platform generates structured IOC exports that can be directly consumed by SIEM platforms, threat intelligence tools, and incident response workflows:

IOC Categories Exported

- Email sender addresses (spoofed/malicious)
- Malicious URLs extracted from the email body
- Suspicious domains with registration data
- VirusTotal detection metadata
- MITRE ATT&CK technique identifiers

Export Formats

PDF Report	Human-readable investigation report with full analysis summary
IOC JSON	Machine-readable structured IOC data for automation pipelines
IOC CSV	Spreadsheet-compatible IOC list for analyst review
Wazuh Alert	Pre-formatted JSON alert compatible with Wazuh SIEM rules

11. MITRE ATT&CK Coverage

The platform automatically maps detected behaviours to MITRE ATT&CK Enterprise techniques:

Technique ID	Name	Description
T1566.002	Spearphishing Link	Attacker sends email with malicious link to steal credentials or deliver malware
T1056	Input Capture	Attacker captures keystrokes or input after victim clicks phishing link
T1656	Impersonation	Attacker impersonates legitimate organization (Binance) to gain trust
T1566	Phishing	Base phishing technique — deceptive email to manipulate recipient action

12. Conclusion

The AI-Powered Phishing & BEC Investigation Platform successfully demonstrates the integration of multiple threat intelligence sources with AI-driven email analysis in a single, analyst-friendly interface. Key achievements include:

- Rapid risk assessment delivering a 0-100 risk score within seconds
- Accurate BEC detection leveraging both keyword heuristics and AI pattern recognition
- Integration with VirusTotal and WHOIS for real-time domain and URL reputation checks
- MITRE ATT&CK mapping for standardized threat communication
- AI-generated natural-language explanations that make findings accessible to non-technical stakeholders
- Multi-format export enabling seamless integration with existing SOC workflows and SIEM platforms

The platform proved highly effective in the test scenario, correctly identifying a Binance impersonation phishing email with a Risk Score of 85/100 and flagging all relevant BEC indicators, malicious URLs, and attack techniques. Future enhancements may include email header parsing, attachment sandboxing, and MISP threat intelligence platform integration.

13. References

- MITRE ATT&CK Framework — <https://attack.mitre.org>
- VirusTotal API Documentation — <https://docs.virustotal.com>
- Anthropic Claude API — <https://docs.anthropic.com>
- Streamlit Documentation — <https://docs.streamlit.io>
- FBI IC3 BEC Report 2024 — <https://www.ic3.gov>
- WHOIS Python Library — <https://pypi.org/project/python-whois>
- Wazuh SIEM Documentation — <https://documentation.wazuh.com>